MILITARY SPECIFICATION

TELETYPEWRITER RELAY CENTRAL AN/MGC-23( )

1. SCOPE

1.1 Scope.-This specification covers a transportable, teletypewriter relay central capable of operation over eight full duplex, direct current (dc) neutral or voice frequency (VF) synchronous secure circuits; designated Teletypewriter Relay Central AN/MGC-23( ). (See 6.5)

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (for Carton Sealing)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-C-3885	Cable Assemblies and Cord Assemblies (for use in Electronic and Associated Electrical Equipment.)
MIL-W-6858	Welding, Aluminum, Magnesium, Non Hardening Steels or Alloys and Titanium, Spot, Seam and Stitch
MIL-T-7928	Terminals Lug and Splice, Crimp Style, Copper
MIL-I-11748	Interference Reduction for Electrical and Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects

FSC 5815

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DRAWINGS

ELECTRONICS COMMAND

SC-DL-506300	Semitrailer, Van, Electronics Facility V-318( )/MGC-23
SC-DL-506317	Teletypewriter Relay Central AN/MGC-23( )
SC-C-506311	Wiring Diagram ZVA Control Box
SC-F-506719	Wiring Diagram, Signal
SC-F-506720	Wiring Diagram, Power
SC-D-82768	Plate, Aircraft Loading Data

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASSOCIATION OF AMERICAN RAILROADS

Rules Governing Loading of Commodities on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren Street, Chicago 5, Illinois.)

3. REQUIREMENTS

3.1 Description.-Teletypewriter Relay Central AN/MGC-23( ) consists of 1 each Semitrailer, Van, Electronic Facility V-318( )/MGC-23 and the following installed major components:

2 each Distributor-Transmitter, Teletypewriter TT-123( )/FG
8 each Distributor-Transmitter, Teletypewriter TT-322( )/GGC-9
4 each Power Supply PP-3131( )/GGG-9
4 each Control, Distributor-Transmitter Teletypewriter C-3873( )/GGC-9
4 each Numbering Unit, Teletypewriter, Message MX-3650( )/GGC-9
22 each Reperforator, TeletypewriterTT-346( )FG
1 each Reperforator-Transmitter, Teletypewriter TT-76( )/GGC
1 each Reperforator-Transmitter, Teletypewriter TT-178( )/FG
9 each Rewinder, Teletypewriter RL-218( )/UG
2 each Teletypewriter TT-98( )/FG
1 each Teletypewriter TT-119( )/FG
10 each Terminal, Telegraph TH-22( )/TG
1 each Telephone Set TA-312( )/PT
16 each Cipher Units

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Teletypewriter Relay Central AN/MGC-23( ) requires 115 VAC, 60 cycle, single phase primary power. Field power can be supplied by Generator Set, Diesel Engine, Trailer Mounted PU-407( )/M or Truck Mounted PU-408( )/M, which is not part of the AN/MGC-23( ).

3.1.1 Semitrailer, Van, Electronic Facility V-318( )/MGC-23.-Semitrailer, Van Electronic Facility V-318( )/MGC-23 is a modified twenty-six-foot, six-ton Mobility Command semitrailer, and contains all brackets, mountings, tables, racks, tie-downs, wiring, and facilities enabling installation and operation of the major components cited in 3.1 above. (See 6.2 (d).)

3.2 Preproduction Samples.-Unless otherwise specified in the invitation for bids and the contract, the contractor shall furnish one preproduction sample of Teletypewriter Relay Central AN/MGC-23( ) for approval. (See 4.3)

3.3 Construction.-

(a) Teletypewriter Relay Central AN/MGC-23( ) shall be constructed in accordance with drawings as listed on SC-DL-506317.

(b) Semitrailer, Van, Electronic Facility V-318( )/MGC-23 shall be constructed in accordance with drawings as listed on SC-DL-506300 which is cited on drawing list in (a) above.

3.4 Electrical installation, facility.-

3.4.1 Power wiring, facility.-The facility power wiring shall be in accordance with the drawings and meet the following requirements:

3.4.1.1 Breakdown voltage, insulation.-Electrical insulation shall withstand a 10-second application of 500 volts A.C. The leakage resistance shall not be less than 10 megohms. The wave form shall be approximately sinusoidal at a frequency not over 100 cycles per second. (See 4.6.1.1.)

3.4.1.2 Overload, circuit breakers.-Each breaker on each of the three distribution panels shall trip within 2 to 5 seconds with 4 times rated load. In each case, the ammeter shall indicate load current. (See 4.6.1.2.)

3.4.1.3 Operation of distribution panel circuits.-All components on all circuits of the distribution panels shall meet continuity and operational tests as follows:

Distribution Panel No. 1- see 4.6.1.3 through 4.6.1.12
Distribution Panel No. 2- see 4.6.1.3 and 4.6.1.13 through 4.6.1.20
Distribution Panel No. 3- see 4.6.1.3 and 4.6.1.21 through 4.6.1.28

3.4.2 Signal wiring, facility.-The facility signal wiring shall be in accordance with the drawings and meet the following requirements:

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3.4.2.1 Continuity between entrance boxes.-Electrical continuity shall exist between the terminals in the 26-pair connectors in the Power and Signal Entrance Box and the corresponding binding posts in the Binding Post Signal Entrance Box. (See 4.6.2.1.)

3.4.2.2 Circuits associated with Black Patch Panel.-All circuits associated with the Black Patch Panel shall meet continuity and operational tests as follows:

VF Secure circuits (16 each)	- see 4.6.2.2 through 4.6.2.4
DC Secure circuits (16 each)	- see 4.6.2.5 through 4.6.2.7
Spare circuits 17 to 25	- see 4.6.2.8
Phone, Incoming Line 26	- see 4.6.2.9
Phone binding post pair circuit	-see 4.6.2.10
Connector Signal No. 2	- see 4.6.2.11
Series Jacks (2 sets, 4 each)	- see 4.6.2.12
Multiple Jacks (2 sets, 4 each)	- see 4.6.2.13

3.4.2.3 Circuits associated with Red Patch Panel.-All circuits associated with the Red Patch Panel shall meet continuity and operational tests as follows:

Secure Circuits, Send	- see 4.6.2.14 and 4.6.2.16
Secure Circuits, Receive, and TD Step	- see 4.6.2.15 and 4.6.2.16
TTY-1; TTY-2; TD Step, Set 1 and 2	- see 4.6.2.17
Send Monitor Reperforators; DC Monitor circuits (8 each)	see 4.6.2.18

3.4.2.4 Continuity of ammeter circuits (Red and Black Patch Panels).-Electrical continuity shall exist in ammeter circuits of the Red and Black Patch Panels. (See 4.6.2.19.)

3.4.2.5 Continuity of ZVA control circuits.-Electrical continuity shall exist in circuits between the ZVA control box and the associated ZVA position jacks. (See 4.6.2.20.)

3.4.3 Cord assemblies, patch and phone.-All patch and phone cord assemblies shall be in accordance with Specification MIL-C-3885.

3.5 System Operation.-With all operating components of the teletypewriter Relay Central AN/MGC-23( ) installed as indicated by the drawings, the equipment shall function in accordance with the following requirements:

3.5.1 Test tape preparation.-The Reperforator-Transmitter Teletypewriter TT-76( )/GGC shall be used to prepare a test tape containing a repeated message utilizing all letters. The message shall be repeated ten times without error in perforation. This tape shall be used in subsequent operation. (See 4.6.3.1.)

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3.5.2 Closed-loop operation of Secure Circuits No. 1 through 8, Send, Receive and Send Monitor.-With all equipments associated with each Secure Circuit connected in closed loop fashion, the test tape of 3.5.1 shall be sent, monitored, and received by the respective equipments without fault. (See 4.6.3.2 and 4.6.3.3.)

3.5.3 Set positions 1 and 2 operation.-With all equipments associated with each Set position patched to a receiver and a send monitor, a message shall be sent, monitored, and received by the respective equipments without fault. (See 4.6.3.4.)

3.5.4 Operation of ZVA circuit equipments.-Equipments associated with the ZVA circuit shall enable a message to be duplicated at each position without fault. (See 4.6.3.5.)

3.6 Vehicular transportation.-The Teletypewriter Relay Central AN/MGC-23() shall be capable of being driven over cross-country terrain by military vehicles without sustaining any permanent damage. (See 4.7.)

3.7 Railroad transportation.-The Teletypewriter Relay Central AN/MGC-23() shall be capable of being loaded, blocked, and braced on a flatcar for shipment by rail in accordance with Association of American Railroads, "Rules Governing Loading of Commodities on Open "Top Cars", Section 6, Figure 67. When so loaded, the AN/MGC-23( ), shall be subjected to and withstand the test of 4.8 without sustaining any permanent damage.

3.8 Fording.-Teletypewriter Relay Central AN/MGC-23( ) shall be capable of being immersed in water to a depth of 60 inches measured from the bottom of the tires for a period of one hour without any leakage of water into the van. (See 4.9.)

3.9 Blackout protection.-The folding vestibule door and the passageway assembly shall provide blackout protection so that with all ceiling lights in the AN/MGC-23( ) operating, no illumination shall be visible during outside darkness from any angle at a distance of 25 feet. (See 4.10.)

3.10 Suppression of radio frequency interference.-The motors in the exhaust blower, air-conditioners and heaters, and the fluorescent lighting used in the van facility shall meet the emanation limits of Specification MIL-I-11748 for Class III. (See 4.4.)

3.11 Mounting Inserts.-All mounting inserts shall be installed without the use of any auxiliary materials such as adhesive and when installed each insert shall be capable of withstanding the torque load referenced in paragraph 4.11. There shall be no damage to the insert threads, the insert shall not have rotated from its original position, and there shall be no distortion of the van mounting surface or insert.

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3.12 Cleaning.-After fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in the applicable portions of Specification MIL-F-14072. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled. After assembly units shall be cleaned thoroughly and shall be free from particles of solder, flux, scale, and any other foreign material. In addition, when necessary, such cleaning shall also be performed before and after final assembly of the units.

3.13 Finish.-The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings.

3.14 Marking.-

3.14.1 General.-Marking shall conform to Specification MIL-M-13231 and the equipment drawings.

3.14.2 Aircraft loading data plate.-The AN/MGC-23( ) shall be provided with a data plate in accordance with the drawings. (For procurement of V-318( )/MGC-23 the plate shall be in accordance with Drawing SC-D-82768.)

3.14.3 Nameplate.-Each AN/MGC-23( ) and each V-318( )/MGC-23 shall be serial numbered and provided with a nameplate in accordance with the drawings

3.14.4 Interior and exterior marking.-All interior and exterior marking of the equipment shall be as depicted on the drawings.

3.15 Riveting.-Riveted joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.16 Securing of parts.-Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall retain components of the AN/MGC-23( ) securely so that loosening, wear of mounting arrangements, or permanent separation of parts or components will not occur when the equipment is subjected to the specified transport tests.

3.17 Soldering.-

3.17.1 Solder and flux.-Solder and flux shall be as defined on the drawings.

3.17.2 Acid or acid salts.-No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

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3.17.3 Process.-There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.18 Tropicalization of material.-Canvas and materials as indicated on the drawings shall be treated to be resistant to moisture and fungi as follows:

3.18.1 Treating materials.-Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.18.2 Toxicity.-Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.18.3 Flexibility.-Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.18.4 Statement of treatment.-The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2.)

3.19 Welding.-Welded joints shall be as defined by the drawings and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858. Class B. All surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from scale, paint, grease and other foreign materials. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slog inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed. (See 4.4.)

3.20 Wiring and cabling.-Wiring and cabling shall be in accordance with the drawings and shall be neat and sturdy.

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3.20.1 Slack.-Wires and cables shall be as short as practical except that sufficient slack shall be provided:

(a) To prevent undue stress on cable forms, wires, and connections.

(b) To facilitate field repair of broken or cut wires.

(c) To prevent chafing or breaking of wires due to repeated flexing of hinged parts.

3.20.2 Protection.-Wires and cables shall be so placed and protected as to avoid contact with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.20.3 Splicing.-Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.20.4 Crimping of solderless terminal lugs.-Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.4 and 4.12.)

3.20.5 Identification of wiring.-Wiring shall be identified by color-coding and wire pair marking, as indicated on the drawings.

3.21 Interchangeability.-Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment. (See 4.13.) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.22 Workmanship.-The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

3.3 Construction
3.12 Cleaning
3.13 Finish
3.14 Marking
3.15 Riveting
3.16 Securing of parts
3.17 Soldering
3.18 Tropicalization
3.19 Welding
3.20 Wiring and cabling

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.-Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery.) (See 4.3.)

(b) Inspection covered by subsidiary documents. (See 4.4.)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5.)

(2) Quality conformance inspection of preparation for delivery. (See 4.15.)

4.3 Preproduction inspection.-This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (see tables III, III, and IV, respectively). The preproduction inspection will normally be performed in this order:

(a) Mounting inserts, (b) Power wiring, facility, (c) Signal wiring, facility, (d) System operation, (e) Vehicular transportation, (f) Rail transportation, (g) Fording. The last tests performed shall be a repeat of the Systems operation tests. The remaining preproduction tests shall be performed in any order satisfactory to the contracting officer.

Table I - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req't Para	Insp Para
Vehicular transportation	3.6	4.7
Rail transportation	3.7	4.8

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4.4 Inspection covered by subsidiary documents.-The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Cord assemblies	3.4.3
Suppression	3.10
Finish	3.13
Marking	3.14
Welding	3.19
Terminals	3.20.4

4.5 Quality conformance inspection of equipment before preparation for delivery.-The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspections procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.6.) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.-This inspection, including sampling, shall conform to Table II and the inspection procedures of Standard MIL-STD-105 using the general inspection levels.

Table II. - Group A inspection

Inspection	Req't Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical	3.22	4.14	1.5 dphu*	6.5 *
Electrical				
Power wiring, continuity, and operation				
Distribution Panel No. 1 circuits	3.4.1.3	4.6.1.3 through 4.6.1.12	1% for the entire group of power	
Distribution Panel No. 2 circuits	3.4.1.3	4.6.1.3 and 4.6.1.13 through 4.6.1.20		
Distribution Panel No. 3 circuits	3.4.1.3	4.6.1.3 and 4.6.1.21 through 4.6.1.28		

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Table II - Group A inspection (Cont'd)

Inspection	Req't Para	Insp Para	AQL Major	AQL Minor
Signal wiring, continuity, and operation				
Continuity between entrance boxes VF Secure Circuits	3.4.2.1 3.4.2.2	4.6.2.1 4.6.2.2 through 4.6.2.4		and
DC Secure Circuits	3.4.2.2	4.6.2.5 through 4.6.2.7		Signal
Spare circuits 17 to 25	3.4.2.2	4.6.2.8		wiring
Phone Incoming Line 26	3.4.2.2	4.6.2.9		
Phone binding post pair circuit	3.4.2.2	4.6.2.10		
Connector Signal No. 2	3.4.2.2	4.6.2.11		
Series Jacks	3.4.2.2	4.6.2.12		and
Multiple Jacks	3.4.2.2	4.6.2.13		
Secure Circuits (Red Patch Panel)	3.4.2.3	4.6.2.14 through 4.6.2.16		System
TTY-1 and 2; TD Step, Set 1 and 2	3.4.2.3	4.6.2.17		
DC Monitor Circuits	3.4.2.3	4.6.2.18		opera-
Ammeter circuits	3.4.2.4	4.6.2.19		tion
ZVA Control circuits	3.4.2.5	4.6.2.20		
System operation				
Test tape preparation	3.5.1	4.6.3.1		
Secure Circuits operation	3.5.2	4.6.3.2 and 4.6.3.3		
Set positions operation	3.5.5	4.6.3.4		
ZVA circuits operation	3.5.4	4.6.3.5		

* Defects per hundred units
** All defects are major

4.5.2 Group B inspection.-This inspection, including sampling, shall conform to Table III and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

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SPRING, LOW PROFILE, WIRE WRAP

4.5.2.1 Group B sampling plans.-The group B sampling plans, for the AQL's listed in table III shall be as follows:

AQL	Inspection level
4.0%	S-4
6.5%	S-4

4.5.2.2 Order of inspection within group B.-Group B inspection shall be performed on the same equipment in the order in which they appear.

Table III. - Group B inspection

Inspection	Req't Para	Insp Para	AQL (see 4.5.2.1)
Mounting insert	3.11	4.11	4.0%
Blackout protection	3.9	4.10	6.5%
Fording	3.8	4.9	6.5%
Breakdown voltage, insulation	3.4.1.1	4.6.1.1	4.0%
Overload, circuit breakers	3.4.1.2	4.6.1.2	4.0%
Interchangeability	3.21	4.13	4.0%

4.5.3 Group C inspection.-This inspection shall be as listed in table IV, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table IV. - Group C inspection

Inspection	Req't Para	Insp Para
Solderless-terminal-lug connection	3.20.4	4.12

4.5.3.1 Sampling for inspection of solderless-terminal-lug connections.-Five specimens of each combination of wire and terminal shall be prepared each week for inspection. A specimen is a 1 - foot length of wire on which a terminal lug has been applied, using the same tools and procedures used in crimping the terminal lugs in production.

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4.5.3.2 Noncompliance.-If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.4.)

4.5.4 Reinspection of conforming group B and group C sample units.-Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Electrical installation tests.-The tests of 4.6.1 and 4.6.2 shall be performed on each van facility (V-318( )/MGC-23) prior to installation of other major components comprising the AN/MGC-23( ). Before applying power to the van facility, an electrical ground at the power and Signal Entrance Box shall be made. Power to Distribution Panels shall be 115 VAC, 60 cycles, single phase.

4.6.1 Power wiring, facility.-Reference Drawing SC-F-506720.

4.6.1.1 Breakdown voltage test.-The equipment shall be tested for conformance to 3.4.1.1. Remove all resistive and inductive loads as follows:

(a) Remove the one-half ampere fuse from the Auxiliary Light Relay circuit and manually activate the relay.

(b) Remove red lead from coil of Entrance buzzer.

(c) Remove cover from Deject Relay Box and disconnect black lead.

(d) Remove all fluorescent lamps from their fixtures.

Turn on breakers on each of the three distribution panels and all associated switches. Close van facility doors and place the By-pass-Blackout switch in Blackout position. Place the Normal-Emergency light switch in Normal position. Apply test voltage between terminals 1 and 4 of the power-in connectors. After test turn off all switches and breakers and restore resistive and inductive loads.

4.6.1.2 Overload test of circuit breakers.-Each breaker shall be tested for conformance to 3.4.1.2 using a dummy load.

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4.6.1.3 Operational tests of Distribution panels.-Continuity between corresponding terminals of Connectors No. 1 and No. 4 (phase A), No. 2 and No. 5 (phase B), and No. 3 and No. 6 (phase C) on the Power and Signal Entrance Box and between terminals of cable assemblies and connectors used for connecting AC power to the van facility shall be verified using a low voltage indication device such as a buzzer (Terminals 1 to 1, 2 to 2, and 4 to 4 on Cable Assembly, Power, Electrical CX-7453( )/U, and 1 to black, 2 to white, and 4 to red on Cable Assembly, Power, Electrical CX-7705( )/U). With power connected, voltmeters on panels shall indicate nominal 115 VAC and frequency meters shall indicate 60 cycle current. The neon lamps shall be lighted. Open and secure vent for exhaust fan and put By-pass-Blackout switch in the By-pass position.

4.6.1.4 Test of Distribution Panel No. 1, Breaker 3 circuit.-

(a) Turn Breaker #3 on, Neon indicator shall light. Turn on Odd and Even light switches. All ceiling lamps shall light. Turn off breaker, lamps shall extinguish; turn breaker on again, lamps shall light. Turn Odd and Even switches alternately off, then on, to indicate that lamps are properly wired to their respective switches.

(b) Close van rear and side doors, turn vestibule light switch on (lamp shall light), close vestibule sliding door and put By-pass-Blackout switch in Blackout position.

Note: Vestibule light shall remain lighted at all times except when extinguished by its own switch.

(c) Open roadside door, all lamps shall extinguish; close door, lamps shall light.

(d) Open curbside door, all lamps shall extinguish; close door, lamps shall light.

(e) Open vestibule door, lamps shall remain lighted. Open rear door, lamps shall be extinguished. Close vestibule door, lamps shall light.

(f) Place By-pass-Blackout switch in By-pass position and place Dome Light switch to the Off position. Using a temporary 24 VDC supply, connect to the emergency wiring of the dome light switch and turn the dome light to Emergency position. Turn main breaker off, the dome lights shall light, all other lamps shall be extinguished..Turn dome light switch to the Off position, dome lights shall be extinguished. Put main breaker on.

(g) Turn dome light switch to Emergency position. Remove ½ amp fuse (for Auxiliary Lights Relay) from Panel No. 1; the dome lights shall light all other lamps shall be extinguished. Replace fuse; dome lights shall be extinguished and all other lamps shall light. Turn dome light switch off and disconnect battery.

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(h) Keeping the By-pass-Blackout switch in By-pass position and the dome light switch in the Off position connect the temporary 24 VDC supply (+) terminal to Pin "K", (-) terminal to Pin "T", to the tractor connector. Place dome light switch to Tractor position and turn off the main breaker, dome lights shall light, all other lamps shall be extinguished. Turn dome light switch to the Off position, dome light shall be extinguished. Put main breaker on and disconnect battery.

Note: By-pass-Blackout switch controls dome lights the same as ceiling lamps.

4.6.1.5 Test of Distribution Panel No. 1, Breaker 1 circuit.-

(a) Before proceeding further open all air conditioning panel covers on the outside of the van.

(b) Turn on Breaker #1. Neon indicator shall light.

(c) Turn Air Conditioner #1 controller to "VENTILATE". Air current shall emit from ducts. Record current indicated on panel #2.

(d) Turn controller to "EXTERNAL" and #1 Thermostatic Control to lowest temperature position. After a period of about 5 minutes the temperature of the air from the air duct shall be cooler than the ambient air in the van. The unit shall function smoothly and the ammeter reading shall have increases above the first observed reading.

(e) Turn off Breaker #1.

4.6.1.6 Test of Distribution Panel No. 1, Breaker 2 circuit.-

(a) Turn on breaker #2. Neon indicator shall light.

(b) Using a 115 VAC test lamp with a plug termination, plug into each of the Cipher 7 and 8 and Cipher Set Pos 1 outlets. The test lamp shall light at each outlet.

(c) Turn off Breaker #2.

4.6.1.7 Test of Distribution Panel No. 1, Breaker 4 circuit.-

Note: This breaker controls the ZVA-TTY outlet and 4 equipment outlets through individual switches with individual indicator lamps. The switches and lamps are located in a separate ZVA Control Box. The equipment outlets are located at positions TT-346( )/FG #1, #2, #3, and #4 in the ZVA rack adjacent to the control box.

(a) Turn on Breaker #4. Neon indicator shall light.

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(b) Plug 115 VAC test lamp into ZVA TTY outlet. Lamp shall light.

(c) With control box switches off, there shall be no power to the four equipment outlets as indicated by the test lamp failure to light when plugged into outlets.

(d) Plug 115 VAC test lamp into position #1 outlet and turn on control switch #1. Test lamp and switch indicator lamp shall light. Turn control switch off. Both lamps shall extinguish. Repeat for positions #2, #3, and #4.

(e) Turn off Breaker #4.

4.6.1.8 Test of Distribution Panel No. 1, Breaker 5 circuit.-

(a) Turn on Breaker #5. Neon indicator shall light.

(b) Plug 115 VAC test lamp into supervisor position outlet and into each of the convenience duplex outlets #3 and #4. The test lamp shall light at each outlet.

(c) Turn off Breaker #5.

4.6.1.9 Test of Distribution Panel No. 1, Breaker 6 circuit.-

(a) Turn on Breaker #6. Neon indicator shall light.

(b) Repeat 4.6.1.8 (b) for convenience outlets #5 and #6.

(c) Turn off Breaker #6.

4.6.1.10 Test of Distribution Panel No. 1, Breaker 7 circuit.-

(a) Turn on Breaker #7. Neon indicator shall light.

(b) Connect the electric heater #3 AC line plug into its proper receptacle. With the heater switch set at "FAN" position the heater shall blow air. After a few minutes at the "HEAT" position (with thermostat turned up) the heater shall blow warm air, and shut off with the switch in the straight up position. Check that thermostat controls heater by placing heater in "HEAT" position and turning thermostat setting to colder temperature setting. When the lower temperature is reached, a click shall occur and the panel ammeter shall drop about 10 amps when the heater shuts off. The fan shall continue to blow.

(c) With the heater in one of its operating positions, turn off breaker #7. The heater shall stop.

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4.6.1.11 Test of Distribution Panel 1, Breaker 8 circuit.-

(a) Turn on Breaker #8. Neon indicator shall light.

(b) Repeat 4.6.1.8 (b) for convenience outlets #9 and #10.

(c) Turn off Breaker #8.

4.6.1.12 Test of Distribution Panel 1, Breaker 9 circuit.-

(a) Turn on Breaker #9. Neon indicator shall light.

(b) Repeat 4.6.1.8 (b) for TT-346( )/FG Pos #5 and #6 and Mon Pos #5 and #6 (one outlet at each Mon Pos designated "TT-346" and the other "RL-218).

(c) The remaining two outlets on this circuit are for TH-22( ) Positions #5 and #6. Power to both of these outlets is controlled by individual switches #5 and #6 which have individual indicator lamps; switches and lamps are located on the TH-22( )/FG Switch Panel. With Switch #5 in the off position, the 115 VAC test lamp shall not light when plugged into outlet #5. Turn switch #5 on. Test lamp and switch indicator lamp shall light. Turn switch #5 off, both lamps shall extinguish.

(d) Repeat (c) above for Switch #6.

(e) Turn off Breaker #9.

4.6.1.13 Test of Distribution Panel No. 2, Breaker 1 circuit.-

(a) Turn on Breaker #1. Neon indicator shall light.

(b) Repeat 4.6.1.5 (c) and (d) for air condition #2.

(c) Turn off Breaker #1.

4.6.1.14 Test of Distribution Panel No. 2, Breaker 2 circuits.-

(a) Turn on Breaker #2. Neon indicator shall light.

(b) Repeat 4.6.1.12 (b), (c), and (d) for TT-345( )/FG Pos #1 and #2, Mon Pos #1 and #2, and TH-22( )/FG Pos #1 and #2.

(c) Turn off Breaker #2.

4.6.1.15 Test of Distribution Panel No. 2, Breaker 3 circuit.-

(a) Turn on Breaker #3. Neon indicator shall light.

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(b) Repeat 4.6.1.12 (b), (c), and (d) for TT-346( )/FG Pos #7 and #8, Mon Pos #7 and #8, and TH-22( )/FG Pos #7 and #8.

(c) Turn off breaker #3.

4.6.1.16 Test of Distribution Panel No. 2, Breaker 4 circuit.-

(a) Turn on Breaker #4. Neon indicator shall light. Turn on Breaker 5, Distribution Panel No. 1.

(b) Plug the 115 VAC test lamp into each of the AN/GGC-9 #1 and #2 outlets. The test lamp shall light at each outlet.

(c) Plug TA-312( )/PT phone into the Phone Jack. This shall activate phone deject relay and cut off power to the outlets. The test lamp shall not light when plugged into each outlet.

(d) One of the two red warning lamps (Security Alarm Indicators) on the AN/GGC-9 racks shall be lighted indication that the phone is plugged in.

(e) With TA-312( )/PT phone still in Phone Jack, place Deject-By-pass switch to by-pass position. Warning light shall be extinguished and power shall be available at the outlets as indicated by the lighting of the test lamp. Re-set By-pass relay and remove phone plug from jack.

(f) Turn off Breaker #4. Test lamp shall not light in any of the outlets.

4.6.1.17 Test of Distribution Panel No. 2, Breaker 5 circuit.-

(a) Turn on Breaker #5, Neon indicator shall light.

(b) Repeat 4.6.1.8 (b) for Cipher 9 through 14 outlets.

(c) Turn off Breaker #5.

4.6.1.18 Test of Distribution Panel No. 2, Breaker 6 circuit.-

(a) Turn on Breaker #6. Neon indicator shall light.

(b) Repeat 4.6.1.10 (b) and (c) for Heater #1.

4.6.1.19 Test of Distribution Panel No. 2, Breaker 7 circuit.-

(a) Turn on Breaker #7. Neon indicator shall light.

(b) Repeat 4.6.1.8 (b) for convenience duplex outlets #1 and #2.

(c) Push "Entrance" and "Mail" buttons to right of rear door, Bell and buzzer, respectively, shall sound.

(d) Turn off Breaker #7.

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4.6.1.20 Test of Distribution Panel No. 2, Breaker 8 circuit.-

(a) Turn on Breaker #8. Neon indicator shall light.

(b) Repeat 4.6.1.8 (b) for outside convenience DuPlex outlet located in Power and Signal Entrance Box.

(c) Turn off Breaker #8 and Main Breaker #2.

4.6.1.21 Test of Distribution Panel No. 3, Breaker 1 circuit.-

(a) Turn on Breaker #1. Neon indicator shall light.

(b) Repeat 4.6.1.5 (c) and (d) for air conditioner #3.

(c) Turn off Breaker #1.

4.6.1.22 Test of Distribution Panel No. 3, Breaker 2 circuit.-

(a) Turn on Breaker #2. Neon indicator shall light.

(b) Repeat 4.6.1.12 (b), (c), and (d) for TT-346( )/FG Pos #3 and #4, Mon Pos #3 and #4, and TH-22( )/FG Pos #3 and #4.

(c) Turn off Breaker #2.

4.6.1.23 Test of Distribution Panel No.#3, Breaker 3 circuit.-

(a) Turn on Breaker #3. Neon indicator shall light.

(b) Repeat 4.6.1.8 (b) for Cipher 1 through 6 outlets.

(c) Turn off Breaker #3.

4.6.1.24 Test of Distribution Panel No. 3, Breaker 4 circuit.-

(a) Turn on Breaker #4. Neon indicator shall light.

(b) Repeat 4.6.1.6 (b) for Cipher 15 and 16 and Cipher Set Pos #2 outlets.

(c) Turn off Breaker #4.

4.6.1.25 Test of Distribution Panel No. 3, Breaker 5 circuit.-

(a) Turn on Breaker #5. Neon indicator shall light. Turn on Breaker 5, Distribution Panel No. 1.

(b) Repeat 4.1.6.16 (b), (c), (d), and (e) for AN/GGC-9 #3 and #4 outlets.

(c) Turn off Breaker #5. Test lamp shall not light in any of the outlets.

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4.6.1.26 Test of Distribution Panel No. 3, Breaker 6 circuit.-

(a) Turn on Breaker #6. Neon indicator shall light.

(b) Repeat 4.6.1.10 (b) and (c) for Heater #4 and its outlet.

4.6.1.27 Test of Distribution Panel No. 3, Breaker 7 circuit.-

(a) Turn on Breaker #7. Neon indicator shall light.

(b) Repeat 4.6.1.10 (b) and (c) for Heater #2 and its outlet.

4.6.1.28 Test of Distribution Panel No. 3, Breaker 8 circuit.-

(a) Turn on Breaker #8. Neon indicator should light.

(b) Turn exhaust fan switch On. Fan shall operate and exhaust air from van. Turn switch Off. Fan shall stop. Repeat 4 times. With fan operating, turn Breaker #8 off, fan shall stop. Turn on Breaker #8. Remove 3 amp fuse from holder, fan shall stop. Replace fuse; fan shall operate. Turn off switch.

(c) The 115 VAC test lamp, when plugged into each convenience duplex outlets #7 and #8, shall light. Turn Breaker #8 Off. Test light shall not illuminate in any of these convenience outlets.

(d) Turn off Main Breaker #3.

4.6.2 Signal wiring, facility.-Reference Drawing SC-F-506719. Continuity and operation of signal wiring circuits shall be verified using a low voltage indicating device (buzzer); the 15-feet-long Cable Stub CX-4760( )/U, which has been verified for continuity between terminals of the connector and the corresponding tinned leads; a Plug PJ-047, referred to as "open plug"; and a Plug PJ-047 whose terminals have been shorted, referred to as "shorted Plug".

4.6.2.1 Continuity between entrance boxes.-

(a) With one lead of the buzzer on terminal 1A of connector Signal No. 1 in the Power and Signal Entrance Box and the other lead on binding post 1A of the top row of binding posts in the Binding Post Signal Entrance Box, the buzzer shall sound. Repeat this test for terminals 1B, 2A, 2B etc. through 26B of connector Signal No. 1 and binding posts 1B, 2A, 2B etc. through 26B respectively in the Binding Post Signal Entrance Box. Buzzer shall sound in all cases.

(b) Repeat (a) for terminals of connector Signal No. 2 and corresponding binding posts in Binding Post Signal Entrance Box.

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3 HOLES .156 DIA THRU (2 PLCS) 4 .1 1 1 2 AL UMP

4.6.2.2 Test of VF Secure Circuit No. 1, Send (Black Patch Panel).-

(a) Connect the cable stub connector to connector Signal No. 1 in the Power and Signal Entrance Box bringing tinned leads through curbside door and connect the buzzer leads to cable stub pair 1A and 1B.

(b) Turn "VF-DC" switch for circuit "Secure 1" on Black Patch Panel to "VF" position.

(c) Plug shorted plug into Line 1 jack; buzzer shall sound. Remove plug.

(d) Plug shorted plug into VF Mon jack; buzzer shall sound. Remove plug.

(e) Plug shorted plug into 4 WS jack at TH-22( )/TG Equipment rack, Secure Circuit No. 1; buzzer shall sound.

(f) With shorted plug still in rack 4WS jack, plug open plug into 4 WS jack on Black Patch Panel; buzzing shall stop.

(g) Remove open plug (buzzer shall buzz) and plug into Line 1 on Black Patch Panel; buzzing shall stop.

(h) Disconnect the buzzer and remove plugs.

(i) Connect buzzer across Secure Circuit No 1, Cipher 1, Send, wire pair terminals located at the Secure Racks.

(j) Plug shorted plug into Equip jack; buzzer shall sound.

(k) With shorted plug still in Equip jack, insert open plug into DC Mon jack; buzzing shall stop. Remove plugs.

(l) Plug shorted plug into rack Send jack; buzzer shall sound.

(m) With shorted plug still in rack Send jack, plug open plug into Send jack on the Black Patch Panel; buzzing shall stop.

(n) Remove plugs and disconnect buzzer.

4.6.2.3 Test of VF Secure Circuit No. 1, Receive (Black Patch Panel.-

(a) With "VF" switch, Secure 1 in "VF" position and stub cable still connected to Signal No. 1 connector. Connect buzzer to wire pair 2A and 2B.

(b) Plug shorted plug into Line 2 jack; buzzer shall sound. Remove plug.

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(c) Plug shorted plug into VF Mon jack; buzzer shall sound. Remove plug.

(d) Plug shorted plug into rack 4WR jack; buzzer shall sound.

(e) With shorted plug still in rack 4WR jack, plug open plug into 4WR jack on the Black Patch Panel; buzzing shall stop.

(f) Remove open plug (buzzer shall sound) and plug into Line 2 on the Black Patch Panel; buzzing shall stop.

(g) Disconnect the buzzer and remove plugs.

(h) Connect buzzer across Secure Circuit No. 1, Cipher #2, Receive, wire pair terminals.

(i) Plug shorted plug into Equip jack; buzzer shall sound.

(j) With shorted plug still in Equip jack, insert open plug into DC Mon jack; buzzing shall stop. Remove plugs.

(k) Plug shorted plug into rack Rec jack; buzzer shall sound.

(l) With shorted plug still in rack Rec jack, plug open plug into Rec jack on the Black Patch Panel; buzzing shall stop.

(m) Remove plugs and disconnect buzzer.

4.6.2.4 Test of VF Secure Circuits No. 2 through 8, Send and Receive.- Test the remaining 14 circuits (beginning with Secure Circuit No. 2, Send; Secure Circuit No. 2, Receive; Secure Circuit No. 3, Send etc. through Secure Circuit No. 8, Receive.) by alternately repeating 4.6.2.2 and 4.6.2.3 using associated cable stub pairs, Black Patch Panel and TH-22( )/TG equipment rack jacks and switch positions and Cipher wire pair terminals.

4.6.2.5 Test of DC Secure Circuit No. 1, Send (Black Patch Panel.-

(a) With "VF-DC" switch, Secure 1 in "VF" position and cable stub still connected to Signal No. 1 connector, connect buzzer to wire pair 1A and 1B.

(b) Connect wire pair terminals, Cipher #1, Send, Secure Circuit No. 1. Buzzer shall not sound.

(c) Place "VF-DC" switch, Secure 1, on Black Patch Panel to "DC" position; buzzer shall sound.

(d) Return switch to "VF" position and disconnect wire pair terminals.

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4.6.2.6 Test of DC Secure Circuit N. 1 Receive (Black Patch Panel).-

(a) With "VF-DC" switch Secure 1, in "VF" position and cable stub still connected to Signal No. 1 connector, connect buzzer to wire pair 2A and 2B.

(b) Connect wire pair terminals, Cipher #2, Receive, Secure Circuit No. 1. Buzzer shall not sound.

(c) Place "VF-DC" switch, Secure 1, on Black Patch Panel to "DC" position. Buzzer shall sound.

(d) Return switch to "VF" position and disconnect wire pair terminals.

4.6.2.7 Test of DC* Secure Circuits No. 2 through 8, Send and Receive.-

Test the remaining 14 circuits(beginning with Secure Circuit No. 2, Send; Secure Circuit No. 2, Receive; Secure Circuit No. 3, Send etc. through Secure Circuit No. 8 Receive.) by alternately repeating 4.6.2.5 and 4.6.2.6 using the associated cable stub pairs, "VF-DC" switches and Cipher wire pair terminals.

*Note: Placing "VF-DC" switch to "DC" position, by-passes TH-22( )/FG equipments.

4.6.2.8 Test of Spare Circuits 17 to 25 (Black Patch Panel).-

(a) With cable stub still connected to Signal No. 1 connector, connect buzzer to pair 17A and 17B.

(b) Plug shorted plug into Line 17 jack, Spare Pairs From Signal Connector No. 1, on the Black Patch Panel. The buzzer shall sound.

(c) Repeat (a) and (b) for spare circuits 18 through 25 using associated cable stub pairs and jacks.

(d) Remove plug and disconnect buzzer.

4.6.2.9 Test of Phone, Incoming Line 26 circuit (Black Patch Panel).-

(a) Telephone Set TA-312( )/PT at Supervisor's Position in the van shall be unplugged.

(b) Using hook-up wire, connect second telephone to binding post pair 26A and 26B of the binding post Signal Entrance Box(in center row).

(c) Using hand crank of phone connected to binding posts, " "ring-up" supervisor's position. Ringer on wall at Supervisor's Position shall sound. Plug in supervisor's phone and determine that conversation is established between phones.

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(d) Put open plug into Phone-Incoming Line 26 jack on Black Patch Panel; phone conversation shall be interrupted. Remove plug.

(e) Hang up both phones and using handcranks verify that either phone rings-up the other. With Supervisor's phone plugged in, the phone shall ring; the wall mounted ringer shall not.

(f) Patch from Phone Incoming Line 26 to Spare Pair Line 17 on the Black Patch Panel using a patch cord with a Plug PJ 047 at each end.

(g) Connect the second phone to binding post pair 17A and 17B and "Ring-up" supervisor's phone. Determine that conversation is established.

(h) Repeat (f) and (g) above for Spare Pair Lines 18 through 25 using associated binding posts.

(i) Disconnect phones and patch cord.

4.6.2.10 Tests of Phone Binding Post circuit.-

(a) Repeat test of 4.6.2.9 (a), (b), (c) and (e) except second phone shall be connected to Phone Binding Post pair on the Power and Signal Entrance Box.

(b) Disconnect phones.

4.6.2.11 Continuity between connector Signal No. 2 and Black Patch Panel.-

(a) Connect the cable stub connector to connector Signal No. 2 in the Power and Signal Entrance Box and connect the buzzer to leads 1A and 1B of the cable stub.

(b) Plug shorted plug into Line 1 jack, Pairs From Signal Connector No. 2, on the Black Patch Panel. Buzzer shall sound.

(c) Repeat (b) above for Lines 2 through 26 with buzzer across associated leads 2A and 2B through 26A and 26B.

(d) Remove shorted plug, buzzer, and cable stub.

4.6.2.12 Test of Series Jacks on Black Patch Panel.-

(a) Using patch cord with a Plug PJ-047 at each end place buzzer across tip and sleeve of one plug and put other plug into Series Jack 1; buzzer shall sound.

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(b) Repeat for Series Jacks 2 through 8 inclusive.

(c) Remove cord.

4.6.2.13 Test of Multiple Jacks on Black Patch Panel.-

(a) Using Patch cord with a Plug PJ-047 at each end, place buzzer across tip and sleeve of one plug and insert other plug into Multiple Jack 1, buzzer shall not sound. Repeat for Multiple Jacks 2 through 8.

(b) With buzzer plugged into Jack 1, insert shorted plug into Jack 2. Buzzer shall sound. Move shorted plug to Jacks 3 and 4. Buzzer shall sound in each case.

(c) Repeat (b) for second set of Multiple Jacks using buzzer in Jack 5, and shorted plug in jacks 6, 7, and 8.

(d) Remove shorted plug, buzzer, and cord.

4.6.2.14 Test of Secure Circuit No. 1, Send (Red Patch Panel).-

(a) Remove cover from Terminal Board Box behind Rack 1, Secure 1 and 2, exposing Terminal Boards TB-1, 2, and 16.

(b) Connect buzzer across Send terminals of TB-1 and insert shorted plug into Ciph 1 jack. Buzzer shall sound.

(c) With shorted plug still in Ciph 1 jack, insert Plug PJ-309 (special patch cord for red DC Mon jack has PJ-309 on each end) into Red DC Mon jack. Buzzing shall stop. Remove plugs.

(d) At the AN/GGC-9( ), Pos 1 rack, connect TD Sig 1 wire pair. Buzzer shall sound.

(e) Insert open plug into TD Sig jack on Red Patch Panel. Buzzing shall stop.

(f) Separate TD Sig 1 wires, remove plug and buzzer.

4.6.1.15 Test of Secure Circuit No. 1, Receive and TD Step (Red Patch Panel).-

(a) With cover to terminal Board Box behind Rack 1 removed, connect buzzer across Rec terminals of TB-1. Buzzer shall sound.

(b) Insert, open plug into the Rec jack of Pos 1 at the TT-346( )/FG racks. Buzzer shall stop. Remove plug; buzzer shall sound.

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(c) Insert open plug into Reperf jack on Red Patch Panel. Buzzing shall stop. Remove plug. Buzzer shall sound.

(d) Insert open plug into DC Mon jack. Buzzing shall stop. Remove plug. Buzzer shall sound.

(e) Reinsert open plug into Reperf jack (buzzing shall stop) and insert shorted plug into Ciph 2 jack. Buzzer shall sound.

(f) Remove plugs and buzzer.

(g) Connect buzzer across Cipher 1 terminals of TB-16 and insert shorted plug into Ciph 1 jack, TD Step, on Red Patch Panel. Buzzer shall sound. Remove plug.

(h) At the AN/GGC-9( ), Pos 1 rack, connect TD Step 1 wire pair. Buzzer shall sound.

(i) Insert shorted plug into AN/GGC-9 jack, TD Step, on Red Patch Panel. Buzzing shall stop.

(j) Separate TD Step 1 wires, remove plug and buzzer.

4.6.2.16 Test of Secure Circuits No. 2 through 8, Send, Receive and TD Step (Red Patch Panel).-

(a) Remove covers from Terminal Board Boxes behind the three remaining Secure Racks exposing Terminal Boards.

(b) Test the remaining 14 circuits (beginning with Secure Circuit No. 2, Send; Secure Circuit No. 2, Receive and TD Step; Secure Circuit No. 3, Send etc. through Secure Circuit No. 8 send, Receive and TD Step) by alternately repeating 4.6.2.14 and 4.6.2.15 using the associated Terminal Boards, AN/GGC-9( ) rack wire pairs, and TT-346( )/FG Rack, and Red Patch Panel jacks.

(c) Do not replace covers of Terminal Board Boxes.

4.6.2.17 Test of Set TTY-1 and 2 circuits (Red Patch Panel.)-

(a) Using a patch cord with a Plug PJ-047 at each end, conne the buzzer across tip and sleeve of one plug and insert other plug into TD a KYBD jack at Set TTY-1. Buzzer shall sound.

(b) Insert open plug into TD Sig jack of Set Teletypewriter No. 1 jack panel located beneath the TT-123( )/FG. Buzzing shall stop. Remove plug. Buzzer shall sound.

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(c) Insert open plug into KYBD jack of set teletype No. 1 jack panel. Buzzing shall stop. Remove plug and patch cord with buzzer.

(d) Plug patch cord with buzzer into PTR jack, Set TTY-1, on Red Patch Panel. Buzzer shall not sound. Insert shorted plug into PTR jack of Set Teletypewriter No. 1 jack panel. Buzzer shall sound. Remove plug and patch cord with buzzer.

(e) Repeat (a), (b), (c), and (d) above for Set TTY-2 jacks on Red Patch Panel and associated jacks on Set Teletypewriter No. 2 jack panel.

4.6.2.18 Test of DC Monitor circuits (Red Patch Panel).-

(a) Connect buzzer across Mon terminals of TB-1 in terminal Board Box behind Secure Rack 1. Buzzer shall sound.

(b) Insert, open plug into Send-Mon jack, Pos 1, at the TT-346( )/FG racks. Buzzing shall stop. Remove plug. Buzzer shall sound.

(c) Insert open plug into DC Moniter Circuit 1 jack on Red Patch Panel. Buzzing shall stop.

(d) Remove buzzer and plug.

(e) Repeat (a) through (d) above for DC Moniter Circuits 2 through 8 using associated terminals boards (referenced in 4.6.2.16 (a)), and jacks on Red Patch Panel and on TT-346( )/FG racks.

(f) Replace covers of Terminal Board Boxes if System operation tests (4.6.3) do not follow.

4.6.2.19 Test of Ammeter Circuits (Red and Black Patch Panels.).-

(a) Disconnect leads from Microammeter at back of Red Patch Panel and twist them together.

(b) Using a patch cord with Plug PJ-039 at each end, connect the buzzer across tip to sleeve of one plug and insert other plug into Micro-amp jack on Red Patch Panel. Buzzer shall sound.

(c) Remove plug with buzzer and reconnect meter leads.

(d) Repeat (a), (b) and (c) for Milliammeter on Red Patch Panel using a patch cord with Plug PJ-047 on each end and the Milliamp jack.

(e) Repeat (d) for Milliammeter on Black Patch Panel using meter MA, jack on Black Patch Panel.

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4.6.2.20 Test of ZVA Control circuits.-Reference Drawing SC-C-506311

(a) Turn ZVA Control Box DC Mon switch to Off position to by-pass milliammeter.

(b) Connect buzzer across leads coming from control box. (These go to Reperforator-Transmitter-Teletypewriter TT-178( )/FG.) Buzzer shall sound.

(c) Insert open plug into ZVA No. 1 jack on adjacent rack; buzzing shall stop; remove plug, buzzer shall sound.

(d) Repeat for ZVA No. 2, No. 3, and No. 4 jacks.

(e) Remove buzzer and plug.

4.6.3 System operation.-

4.6.3.1 Test tape preparation.-The test tape shall be prepared in accordance with 3.5.1. The first two messages shall be typed and the other eight messages shall be duplicates using the transmitter-distributor.

4.6.3.2 Secure Circuit No. 1 operation.-

(a) All circuit breakers shall be in On position.

(b) Turn "VF-DC" switch for Secure Circuit No. 1 on Black Patch Panel to "VF" position.

(c) Remove cover from Terminal Board Box behind Rack 1, Secure 1 and 2, exposing Terminal Boards TB-1, 2, and 16.

(d) Connect Secure Circuit No. 1, Cipher 1, Send wire pair terminals to Send terminals of TB 1.

(e) Observing color coding to maintain polarity, connect one wire of the Cipher 2 Receive pair to one of the REC terminals of TB 1 and connect the other wire of the pair to one of the Mon terminals of TB 1. Connect the unused Rec and Mon terminals of TB 1 with a jumper.

(f) Place jumper across Ciph 1, TD Step terminals of TB 16.

(g) At the AN/GGC-9 rack, Pos 1 verify that TD Sig 1 wire pair is connected to TD Sig 1 terminals on the back of C-3873( )/GGC-9 control, and similarly, that the the TD Step 1 wire pair is connected to the TD Step 1 terminals. Plug in power cord into associated outlet.

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(h) At the Binding Post Signal Entrance Box, jumper binding posts 1A to 2A and 1B to 2B.

(i) Connect TH-22( )/TG Secure Circuit No. 1 jacks to TH-22( )/TG Pos 1 unit with patch cords: Send jack to either Send jack; Rec jack to Rec jack, 4WS jack to 4WS terminals; 4WR jack, to 4WR terminals.

(j) Plug power cord of TH-22( )/TG Pos 1 unit into associated power outlet and put toggle switch for unit into On position.

(k) Insert plug of TT-346( )/FG Receive, Pos 1 unit into Rec jack of Pos 1, and plug power cord of unit into associated outlet. Install a tape roll and turn unit On.

(l) Insert plug of TT-346( )/FG, Send-Monitor, Pos 1 unit into Send-Mon jack of Pos 1, and plug power cord of unit into associated outlet. Install a tape roll and turn unit On.

(m) Turn TT-322( )/GGC-9, Send Ckt 1 unit On. Insert the test tape into the equipment and transmit.

(n) The message shall be received at the Receive TT-346( )/TG and at the Send-Monitor TT-346( )/TG. The message number that appears on the MX-3650( )/GGC-9 numbering unit shall appear at the start of each of the received messages.

(o) As the tape feeds from the Send-Monitor unit, insert the starting end into the adjacent RL-2180/G tape rewinder reel after threading tape over rollers. Turn rewinder on. Rewinder shall take-up tape as it feeds.

4.6.3.3 Operation of Secure Circuits No. 2 through 8.-Repeat test procedure of 4.6.3.2 for the remaining 7 Secure Circuits using the associated switches, terminal boards, cable pairs, AN/GGC-9( ) equipments, binding posts. TH-22( )/TG equipments, TT-346( )/TG Receive and Send-Monitor equipments and rewinders.

4.6.3.4 Set positions operation.-With all equipments connected as they were at the end of testing of Secure Circuit No. 8(4.6.3.3) proceed as follows:

(a) Insert associated plugs of TT-98( )/FG Teletypewriter, Set 1 into TD Sig jack, KYBD jack, and PTR jack at Set Teletypewriter No. 1 jack panel.

(b) Insert a patch cord on Red Patch Panel from Set TTY-1, TD and KYBD jack to the Send, Cipher jack of any of the eight Secure Circuits and a patch cord from Set TTY-1 PTR jack to the Receive, Cipher jack of the same Secure Circuit.

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(c) Turn the TT-98( )/FG on and using the Keyboard send the same message that is used on the test tape. Page copy shall be received.

(d) The message shall be received at the selected Send-Monitor position.

(e) Turn On TT-123( )/FG unit and insert test tape and send message.

(f) The message shall be received as in (d) above.

(g) Repeat (a) through (f) above for Set 2.

(h) Remove all jumpers; disconnect wire pairs, and replace Terminal Board Box covers.

4.6.3.5 ZVA Position operation.-

(a) Interconnect the TT-178( )/FG and TT-119( )/FG equipments at the ZVA Position. Install tape and page rolls.

(b) At ZVA positions 1 through 4 connect power cords of TT-346( )TG units to respective outlets. Insert plugs into respective ZVA jacks. Install tape rolls and turn units On.

(c) Connect wire pair from ZVA Control Box to associated terminals on the TT-178( )/FG unit.

(d) Type message of test tape twice and receive page copy and receive tape at TT-178( )/FG.

(e) At ZVA Control Box, turn on switches to ZVA positions 1 through 4 and Monitor switch. Ammeter shall indicate approximately 20 milliampers.

(f) Feed tape to transmitter-distributor and duplicate message eight additional times. The total message shall be received at each TT-346()/TG unit, ZVA positions 1 through 4.

4.7 Vehicular transportation test.-The Teletypewriter Relay Central AN/MGC-23( ) shall be attached to a Government furnished Truck, Tractor, 5 Ton 6 x 6 M-52, and transported as follows:

(a) On any convenient hard surfaced road it shall be subjected to 60 abrupt stops from 30 miles per hour.

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(b) Over a hard surface having a specially prepared course. The course shall have twelve 4 inch x 8 inch timbers placed 25 feet apart on the 8 inch face and with the 4 inch face fully above the ground. The edges shall have a 1 inch x 1 inch bevel. The sixth and twelfth timbers shall be placed 45 degrees to the direction of travel. The timbers shall be long enough to span the vehicle passing over them and shall be anchored down by any convenient means. The equipment shall experience no damage from traversing the course in either direction, as follows; and after shall meet the tests of 4.6.3:

(1) 10 laps * at 5 mph

(2) 10 laps at 10 mph

(3) 10 laps at 15 mph

(4) 10 laps at 18 to 20 mph

* A lap is defined as traversing the course in one direction.

4.8 Railroad transportation test.-The Teletypewriter Relay Central AN/MGC-23( ) shall be loaded on a railroad flatcar in accordance with 3.7. The test shall be conducted on a flat stretch of track. A loaded car totaling 165,000 lbs, travelling at 9 miles per hour (or equivalent impact energy conditions approved by the Government) shall be impacted against the test flat car coupled to two other empty cars. These cars shall be stationary with brakes off. Four impacts shall be performed, two impacts into each end of the test flat car. The cabling and blocking holding the van in place shall be tight at the start of each impact run. If the cabling or blocking is torn loose by the impact, the run shall be repeated. A damage survey shall be made after each impact. The AN/MGC-23( ) shall experience no permanent deformation during this test and after shall meet the tests of 4.6.3.

4.9 Fording test.-The Teletypewriter, Relay Center shall be tested in accordance with 3.8. No special fording kits shall be used.

4.10 Blackout protection.-The Teletypewriter Relay Central shall be tested for conformance to 3.9. The passageway components shall be assembled and attached to each side door of the van facility. The other end of the passageway assembly shall be supported by auxiliary blocking.

4.11 Mounting insert test.-Four of each size insert shall be randomly selected for test in each sample van facility. Each selected insert shall have a torque load, in accordance with Table V applied to the bolt. (Note: Bolt shall hold a stationary plate against head of insert). After test, the requirements of 3.11 shall be met. Failure of any insert shall be cause for rejection of sample van facility.

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Table V. - Production Torque Load

Insert size	Torque (in lb)
3/8 - 16	150
5/16- 18	100
1/4 - 20	60

4.12 Testing of solderless terminal-lug connections.-Connections made with solderless terminal-lugs shall be tested for conformance to 3.20.4.

4.13 Inspection for interchangeability.-The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.21. When a dimension is not within specified or design limits, it shall be considered a defect.

(a) Size and mounting dimensions of brackets and mounting plates.

(b) Size and mounting dimensions of racks and cabinets.

(c) Size and mounting dimensions of equipment drawers.

(d) Size, mounting dimensions, and capacity of ventilating equipme:

4.14 Visual and Mechanical inspection.-/Equipment shall be examined for defects listed in Standard MIL-STD-252, the requirements of 3.22 and Table VI.

Table VI. - Classification of visual and mechanical defects

Classification	Defects
Major	1. Mounting plates missing or mislocated 2. Aircraft loading data plate, instructions or nameplate omitted, incorrect or illegible. 3. Welds--cracked or porous. 4. Equipments or bracketry loose or missing. 5. Drawers, slides, doors etc. inoperative. 6. Power receptacles, switches, jacks, wire ducts or other electrical components loose, improperly located or inoperative. 7. Locking or holding devices missing or inoperative.

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Table VI. - Classification of visual and mechanical defects (cont'd)

Classification	Defects
	8. Canvas components of passageway assembly have holes, rip or tears.
	9. Gaps between sections of wire duct exceed 1/16".
	10. Folding vestibule door inoperative.
	11. Passageway components connot be assembled.
Minor	1. Drawers, slides, door, not easily operable.
	2. Fastening devices difficult to operate.
	3. Finish - abrasions or scratches.
	4. Passageway components difficult to assemble.

4.15 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.16 Rough handling test (preparation for delivery).-When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.2(c)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Paragraph 4.6.3	System operation (AN/MGC-23( ) procurement)
Paragraphs 4.6.1 and 4.6.2	Power and Signal wiring (V-318( )/MGC-23 procurement)
Paragraph 4.14	Visual and mechanical

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-AN/MGC-23( ) shall be preserved and packaged in accordance with procedures specified for the designated methods prescribed in Specification MIL-P-116, as described in the following paragraphs.

5.1.1.1 Technical Literature.-Technical literature shall be packaged Method 1C-6, as follows: Inclose each technical literature within a close fitting bag, type II, Class d, conforming to Specification MIL-B-117. Heat seal closure.

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5.1.1.2 Wire or cables.-Wind cable on the assigned reels. Secure cable ends to the reels.

5.1.1.3 Stow all components within the designated compartments of the cabinets or drawers of the trailers. Cushioning voids within drawers to prevent movements and damage to components.

5.1.1.4 Consolidate packages.-All loose components not having an assigned place for stowage, shall be consolidated within fiberboard or nailed wood boxes, and shall be properly blocked on the floor of the trailers.

5.1.1.5 Blocking and tying.-All wall hanging instruments shall be reinforced in their position with fiberglass or flat metal strapping.

5.1.2 Level C.-AN/MGC-23( ) components shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.-

5.2.1 Level A.-Secure windows, doors, blower vents, heater or intake doors, louvers or slots. For added water-proof protection, tape the closures with water resistant pressure sensitive tape PPP-T-76.

5.2.2 Level C.-AN/MGC-23( ) shall be packed in a manner that will afford adequate protection against damage to the equipment and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking.-Interior packages and exterior markings shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.-The Teletypewriter Relay Central AN/MGC-23( ) is intended for use with an associated teletypewriter terminal (such as the AN/MGC-22( ) or AN/MGC-32( )) which provides tape preparation facilities for message origination, page printer facilities for message termination, and facilities for communications with satellite subscriber stations.

6.2 Ordering data.-Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

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(b) The specific paragraphs of Section 5 which are applicable to the particular procurement.

(c) Preproduction Pack(s) as follows:

(1) Makeup of pack(s)
(2) Number of each kind of pack to be submitted.
(3) Inspection to be performed thereon--including rough handling test, which will not be performed as procurement inspection.
(4) Five copies of process sheet, using format shown on Drawing SC-C-33073 and accompanied by such drawings, sketches, and other data as may be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the contracting officer with the preproduction pack(s).
(5) A packaging technician from the procuring agency will evaluate the preproduction pack(s).

(d) Procurement of Semitrailer, Van, Electronic Facility V-318( )/MGC-23 should be in accordance with this specification. Preproduction inspection should be performed using Government furnished live-loads for testing purposes unless otherwise specified in bid request or contract.

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) List of each individual component item to be used for testing the equipment.

(h) Submission of the statement of treatment referenced in 3.18.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Inspection.-Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.1 Examination.-Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2 Testing.-Testing consists of determinations of compliance, using technical means.

6.4 Group C inspection.-Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2).

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6.5 Nomenclature.-The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: AN/MGC-23 Z. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification inspection.-Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:
Army (EL)

Preparing Activity:
Army (EL)

Project Number 5815-A027

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No 119-R0

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of pr ducts for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which wi insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will b appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

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